Aayush Mishra

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Efficient and reliable AI: activation steering, adaptive computation, parameter-efficient adaptation

EDUCATION

Johns Hopkins University Ph.D., Computer Science advised by Anqi Liu — Jan 2022 – present
M.S.E., Computer Science | GPA 4.0/4.0 (transcript) — Dec 2023

IIT Mandi B.Tech., Computer Science & Engineering, minor in Management — Aug 2015 – Jun 2019
GPA 8.88/10 (transcript)

EXPERIENCE

MATS Research Fellow (11.0), OpenAI Safety Team Berkeley, CA — Sep 2026 – present

- Researching *natural* activation steering for reasoning and looped LMs, extending my result that standard steering drives models into internal states no prompt can reach.

Qualcomm AI Research AI Research Intern Amsterdam, NL — May – Sep 2026

- Made early-exit computation practical in Looped LMs (Ouro): a best-available KV-caching strategy plus an effort-aware exit objective cut FLOPs and KV-cache memory by up to 30% at full-depth quality.
- Filed 3 invention disclosures; first-author paper on the work (arXiv).

Adobe AI Research Intern College Park, MD — May – Aug 2024

- Co-developed EigenLoRA, which recycles trained LoRA adapters into a shared principal subspace: new tasks train up to 100× fewer parameters and multi-adapter inference uses ∼18× less memory, for on-device LLM and diffusion models.

Microsoft Data & Applied Scientist, Bing Shopping Hyderabad, India — Aug 2021 – Jan 2022

- Built a query-to-product-category classifier over ∼18k classes and a relevance metric to improve offer ranking.

Siemens Research Engineer Bengaluru, India — Jul 2019 – Aug 2021

- Developed a CNN explainability and compression algorithm for edge deployment; filed 3 invention disclosures.
- Trained an RL agent to surface edge-case safety violations of self-driving agents in simulation.
- Internships (2017–18): semantic code search with LSA; NLP classifier for Siemens' internal social network.

SELECTED PUBLICATIONS

Enabling Dynamic Computation in Looped LMs 2026
Aayush Mishra, Arnau Padrés Masdemont, et al. | arXiv

Steered LLM Activations are Non-Surjective 2026
Aayush Mishra, Daniel Khashabi, Anqi Liu | arXiv; Sci4DL, Re-Align @ ICLR

IA2: Alignment with ICL Activations Improves Supervised Fine-Tuning 2026
Aayush Mishra, Daniel Khashabi, Anqi Liu | ICLR

Genomic Next-Token Predictors are In-Context Learners 2026
Nathan Breslow, **Aayush Mishra**, et al. | TMLR; MLGenX (Oral), Sci4DL @ ICLR

ODD: Overlap-aware Estimation of Model Performance under Distribution Shift 2025
Aayush Mishra, Anqi Liu | UAI

EigenLoRA: Recycle Trained Adapters for Resource-Efficient Adaptation and Inference 2024
Aayush Mishra*, Prakhar Kaushik*, et al. | Preprint

Do Pretrained Transformers Learn In-Context by Gradient Descent? 2024
Aayush Mishra*, Lingfeng Shen*, Daniel Khashabi | ICML (Oral)

SKILLS & SERVICE

Tools Python, C++, PyTorch, Hugging Face Transformers/TRL, data engineering

Methods activation steering, interpretability, adaptive computation, KV-cache efficiency, LoRA/PEFT, self-distillation

Reviewer ICML (Silver Reviewer), ICLR, NeurIPS, UAI, TMLR